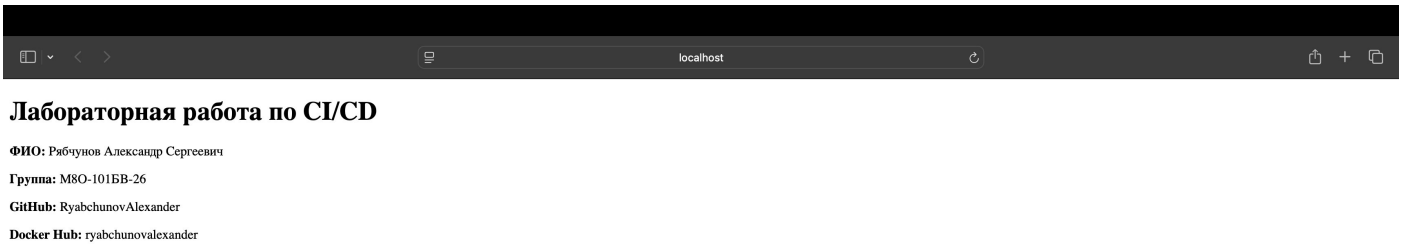
1.1 docker build –t ryabchunovalexander-ci-cd-lab:v1

```
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % docker build -t ryabchunovalexander-ci-cd-lab:v1 .
[+] Building 8.2s (8/8) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 113B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                               8.0s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                                  0.0s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 497B                                                                                             0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:a46f57e097e8effc26192437d2e715adf74bcbea836117651ee436c27983fa41         0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:a46f57e097e8effc26192437d2e715adf74bcbea836117651ee436c27983fa41         0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                                    0.0s
 => exporting to image                                                                                                        0.0s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:a6add26c6cf109619dda3a0f59ec8e132ffbb9e4cc816be6a0b013fa4886f54b                             0.0s
 => => exporting config sha256:0b61cb09a1ed5cf8c5cd1364289bf2a84e63a5b5ab483b230c5fb9c1beaff5df                               0.0s
 => => exporting attestation manifest sha256:276a6a045f40d0773f8e1b241b61c875346c6d80f7ad3a631891320d9d7ef19e                 0.0s
 => => exporting manifest list sha256:10d48d8885549a221a85c29376fbb1509daf068eb623b6804017b225e03ff163                        0.0s
 => => naming to docker.io/library/ryabchunovalexander-ci-cd-lab:v1                                                           0.0s
 => => unpacking to docker.io/library/ryabchunovalexander-ci-cd-lab:v1                                                        0.0s
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab %
```

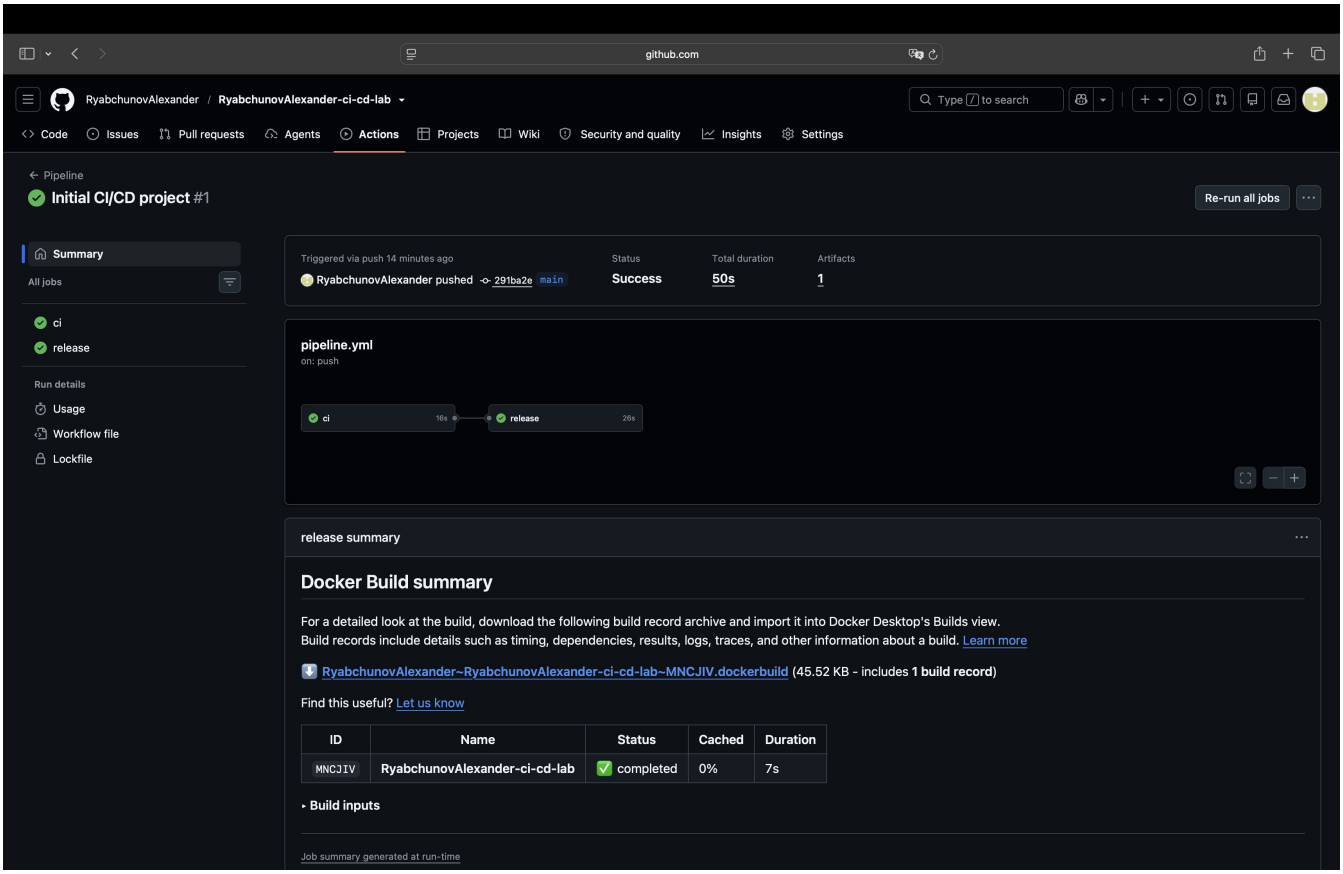
Локальная сборка Docker-образа

1.2 docker run -d --name ci-cd-lab -p 8080:80 ryabchunovalexander-ci-cd-lab:v1

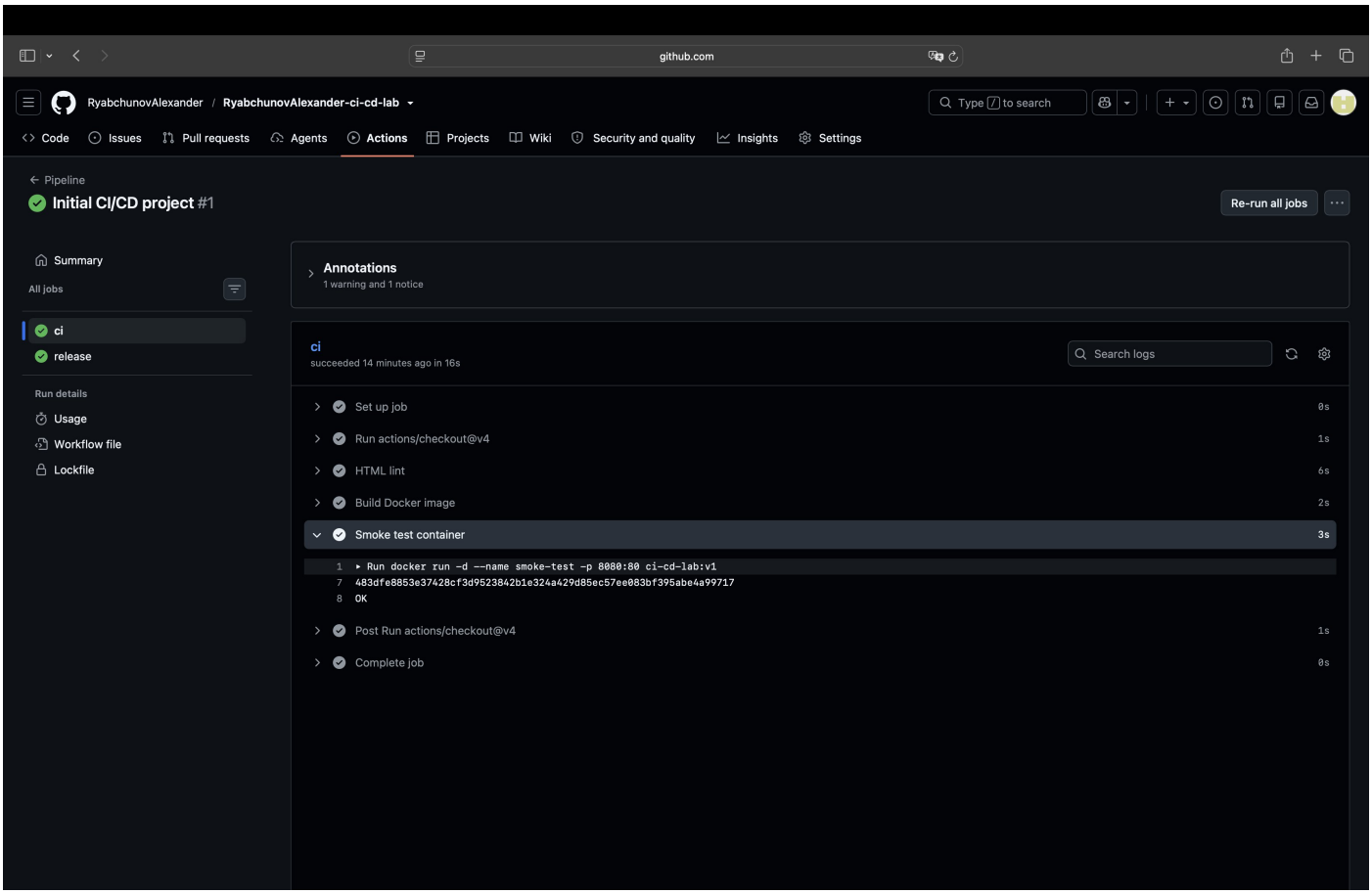


Локальный запуск контейнера и проверка в браузере

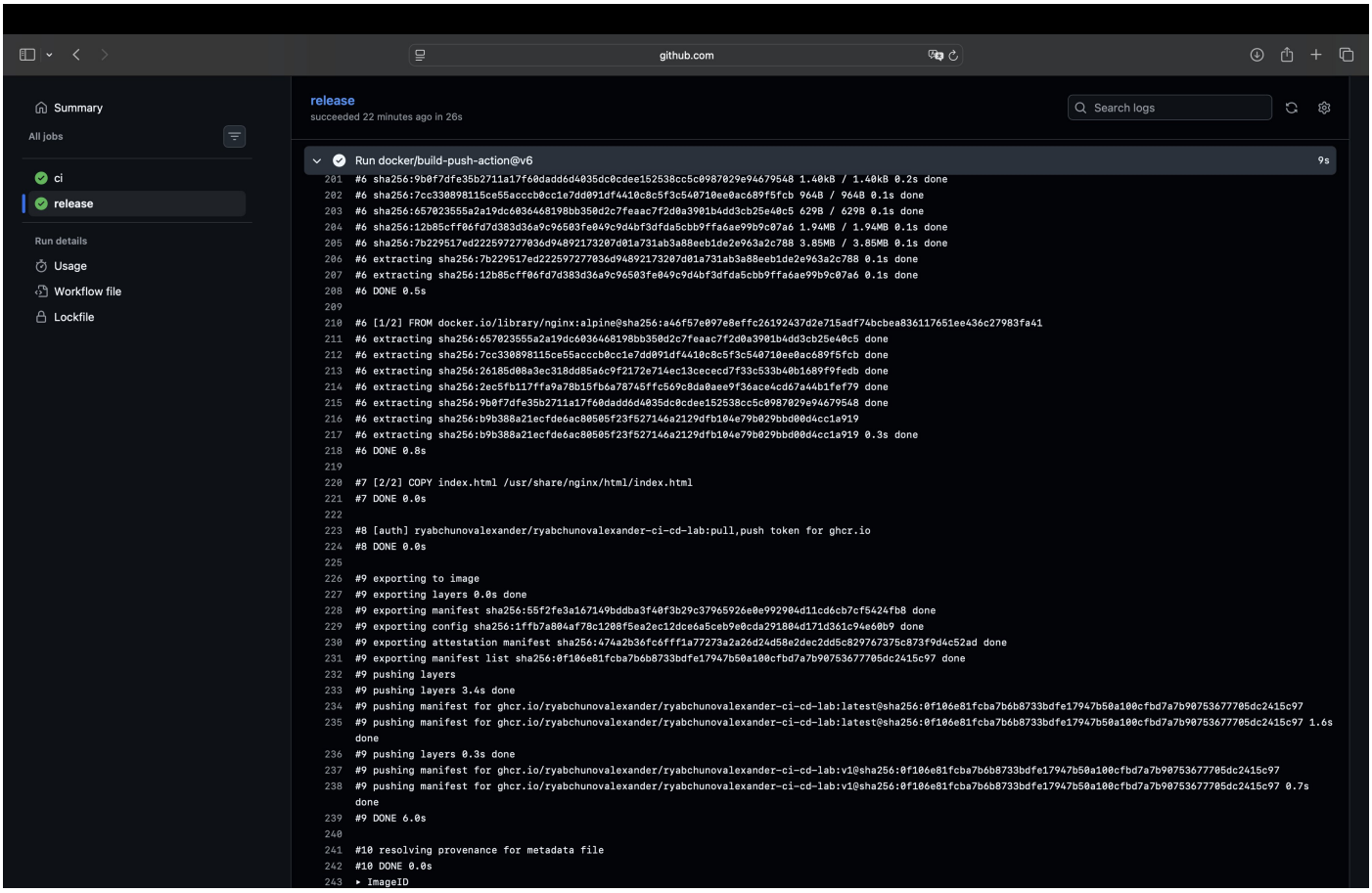
1.3 Проверка выполнения Job ci во вкладке GitHub Actions.



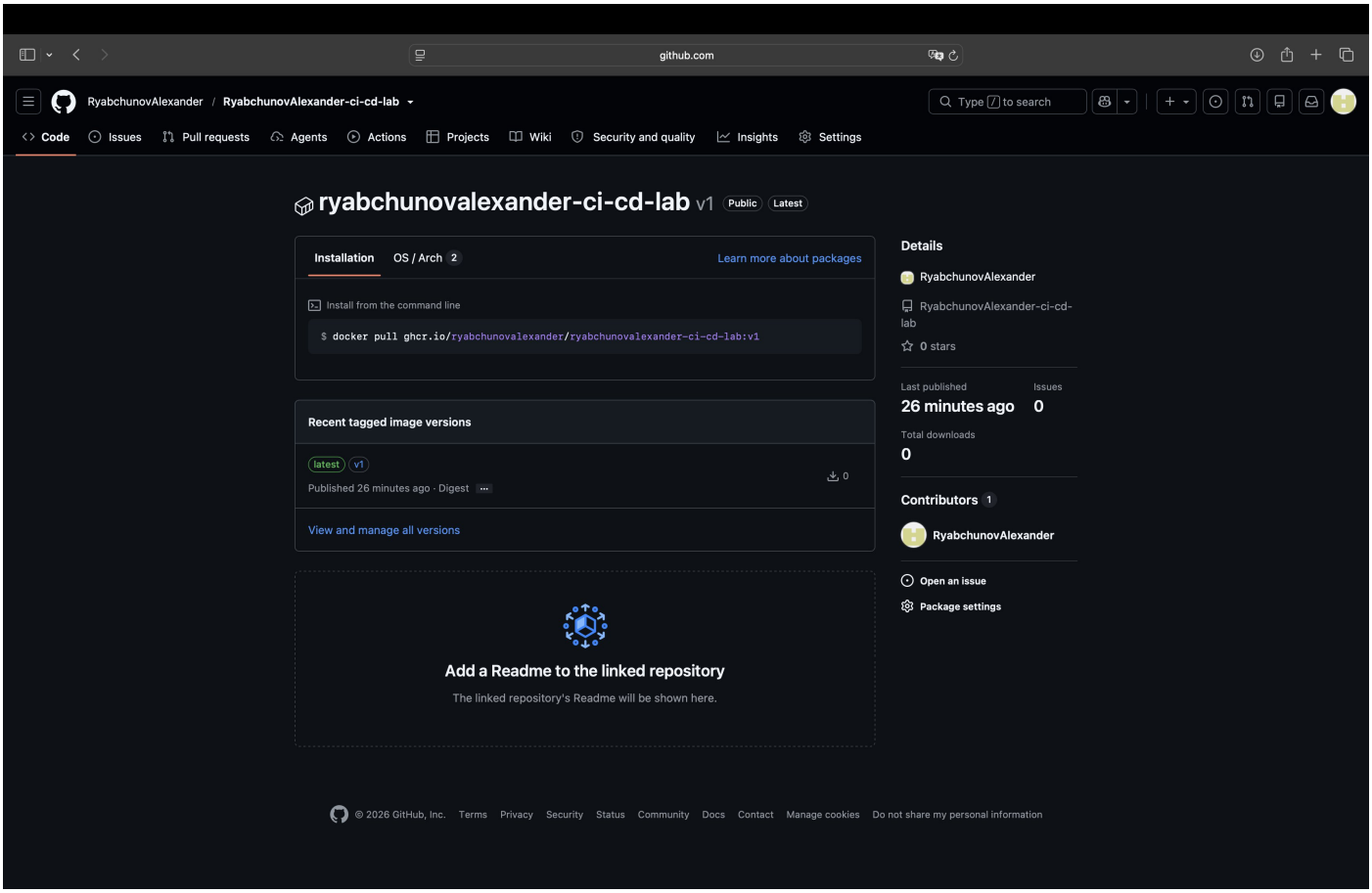
Задание ci выполнено успешно (зеленый статус). В логах шага Smoke test container выведена строка OK



1.4 Проверка выполнения Job release в том же рабочем конвейере



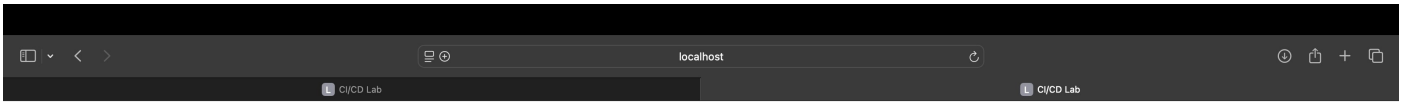
1.5 Проверка публикации пакета в интерфейсе GitHub Packages.



1.6 docker pull ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest

docker run -d --name ghcr-test -p 8081:80 ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest

```
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % docker pull ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest
latest: Pulling from ryabchunovalexander/ryabchunovalexander-ci-cd-lab
7b9d02243238: Already exists
caba72f1dd05: Pull complete
f523b340a6c0: Already exists
dbb1ef19121a: Already exists
567a969e0221: Already exists
c2fefbef5314: Already exists
d3531b7a4f21: Already exists
cbe54b031f3e: Already exists
1408044c7403: Already exists
44136fa355b3: Already exists
eabda052225d: Download complete
Digest: sha256:dd9318ff372f10608918d88a4ccc1481c3c3023c1e6c8e8d1b0025a99610d560
Status: Downloaded newer image for ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest
ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % docker run -d --name ghcr-test -p 8081:80 ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest
3ba0854c83dc2fa70f557daa63e1cc9880796d3e67481c8484f0e83e6cf4e6ae
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab %
```



Лабораторная работа по CI/CD

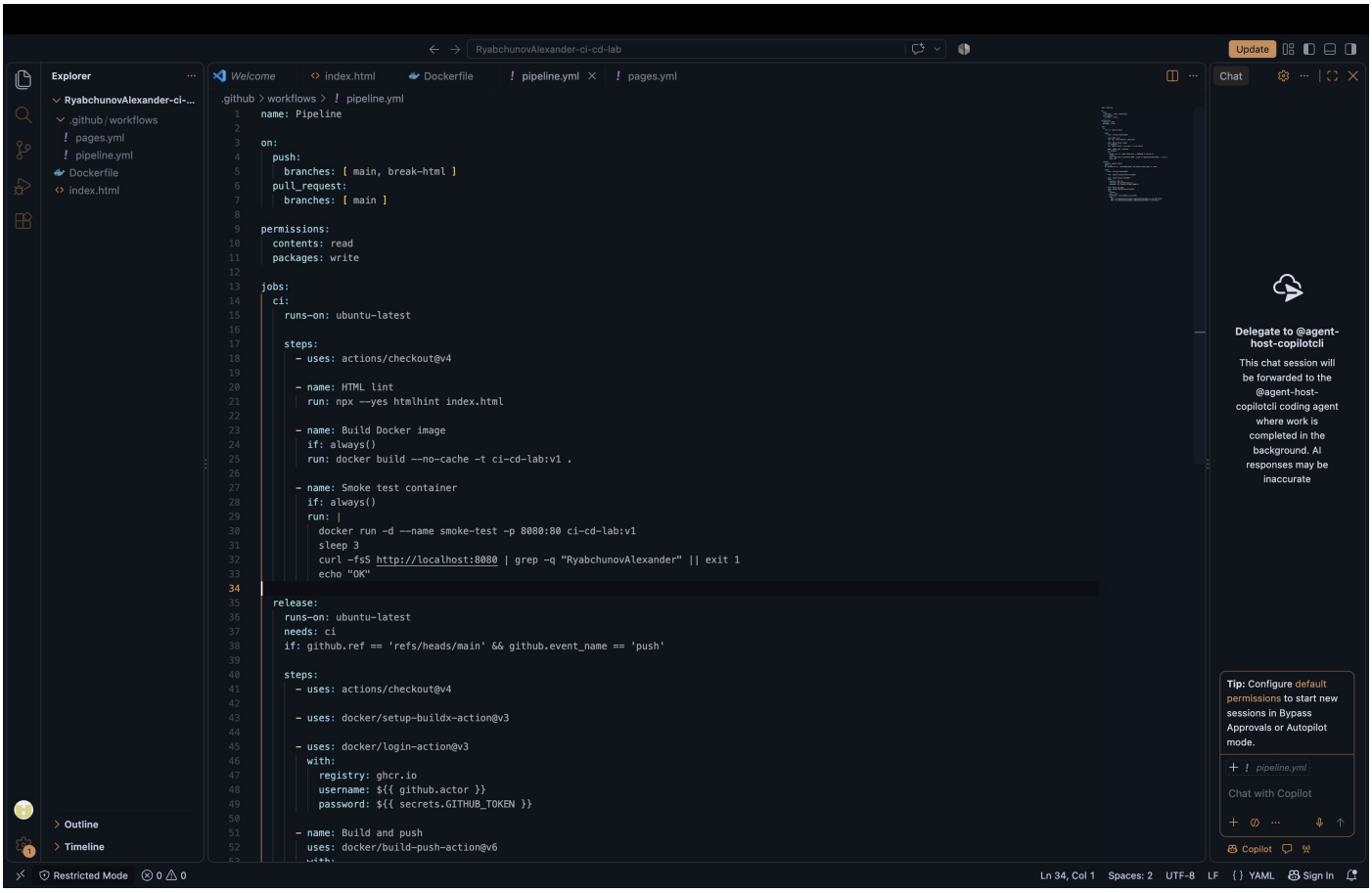
ФИО: Рябчунов Александр Сергеевич

Группа: М8О-101БВ-26

GitHub: RyabchunovAlexander

Docker Hub: ryabchunovalexander

2.1 Демонстрация архитектуры файла конфигурации pipeline.yml.



```yaml
name: Pipeline

on:
  push:
    branches: [ main, break-html ]
  pull_request:
    branches: [ main ]

permissions:
  contents: read
  packages: write

jobs:
  ci:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: HTML lint
        run: npx --yes htmlhint index.html

      - name: Build Docker image
        if: always()
        run: docker build --no-cache -t ci-cd-lab:v1 .

      - name: Smoke test container
        if: always()
        run: |
          docker run -d --name smoke-test -p 8080:80 ci-cd-lab:v1
          sleep 3
          curl -fsS http://localhost:8080 | grep -q "RyabchunovAlexander" || exit 1
          echo "OK"

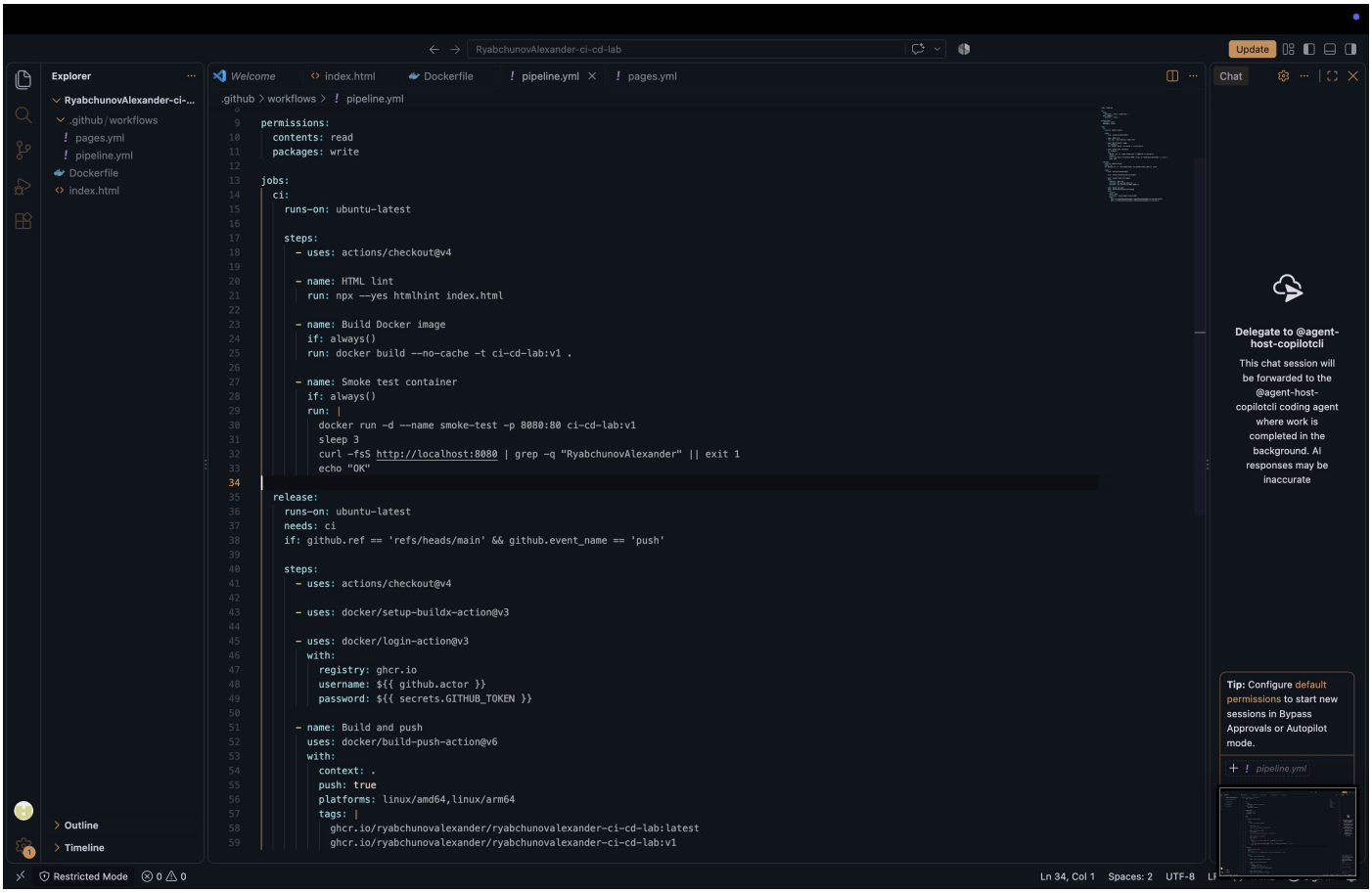
  release:
    runs-on: ubuntu-latest
    needs: ci
    if: github.ref == 'refs/heads/main' && github.event_name == 'push'

    steps:
      - uses: actions/checkout@v4

      - uses: docker/setup-buildx-action@v3

      - uses: docker/login-action@v3
        with:
          registry: ghcr.io
          username: ${{ github.actor }}
          password: ${{ secrets.GITHUB_TOKEN }}

      - name: Build and push
        uses: docker/build-push-action@v6
```



```yaml
permissions:
  contents: read
  packages: write

jobs:
  ci:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: HTML lint
        run: npx --yes htmlhint index.html

      - name: Build Docker image
        if: always()
        run: docker build --no-cache -t ci-cd-lab:v1 .

      - name: Smoke test container
        if: always()
        run: |
          docker run -d --name smoke-test -p 8080:80 ci-cd-lab:v1
          sleep 3
          curl -fsS http://localhost:8080 | grep -q "RyabchunovAlexander" || exit 1
          echo "OK"

  release:
    runs-on: ubuntu-latest
    needs: ci
    if: github.ref == 'refs/heads/main' && github.event_name == 'push'

    steps:
      - uses: actions/checkout@v4

      - uses: docker/setup-buildx-action@v3

      - uses: docker/login-action@v3
        with:
          registry: ghcr.io
          username: ${{ github.actor }}
          password: ${{ secrets.GITHUB_TOKEN }}

      - name: Build and push
        uses: docker/build-push-action@v6
        with:
          context: .
          push: true
          platforms: linux/amd64,linux/arm64
          tags: |
            ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:latest
            ghcr.io/ryabchunovalexander/ryabchunovalexander-ci-cd-lab:v1
```

2.2 git checkout -b break-html

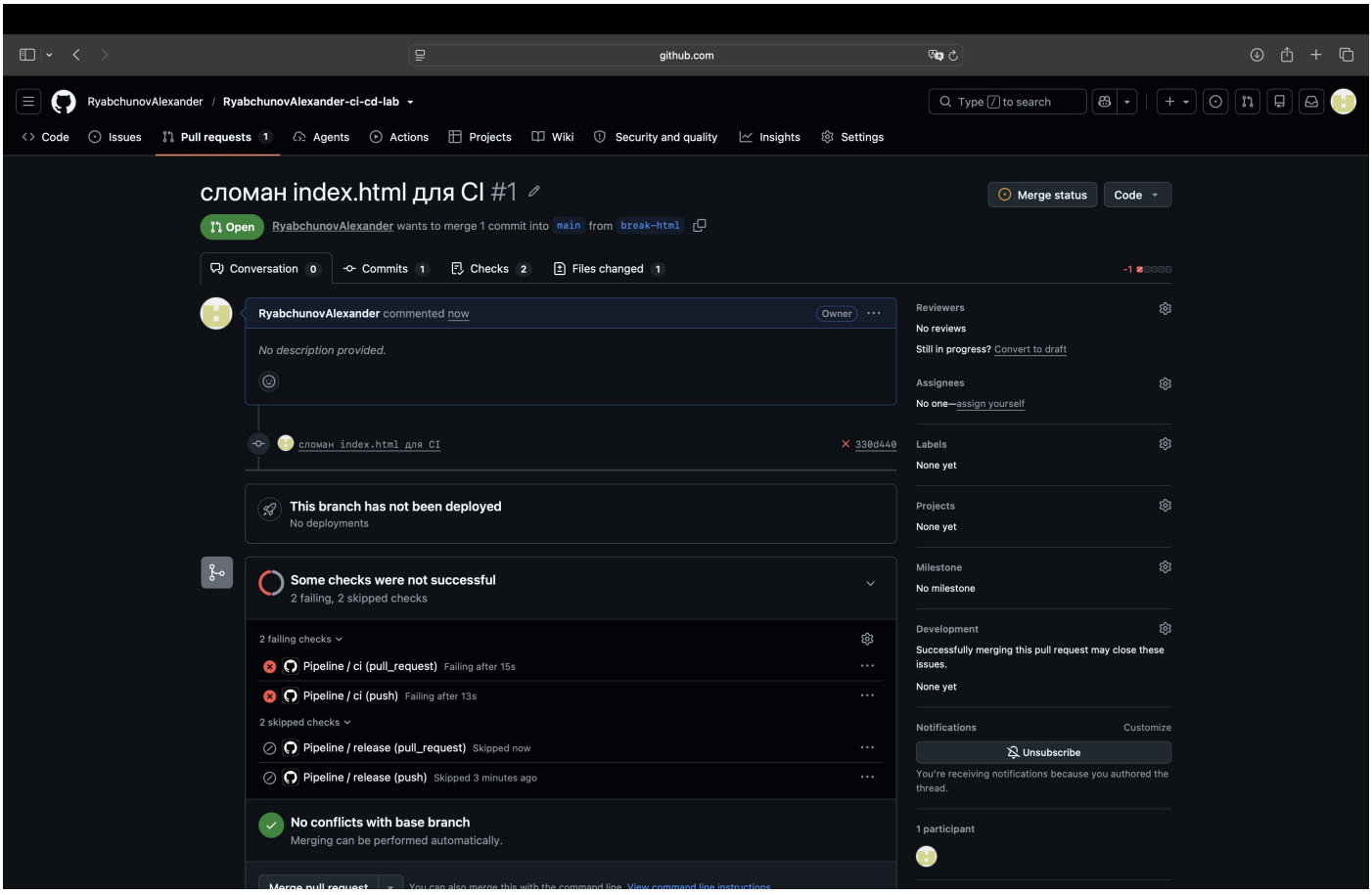
git add index.html

git commit -m "сломан index.html для CI"

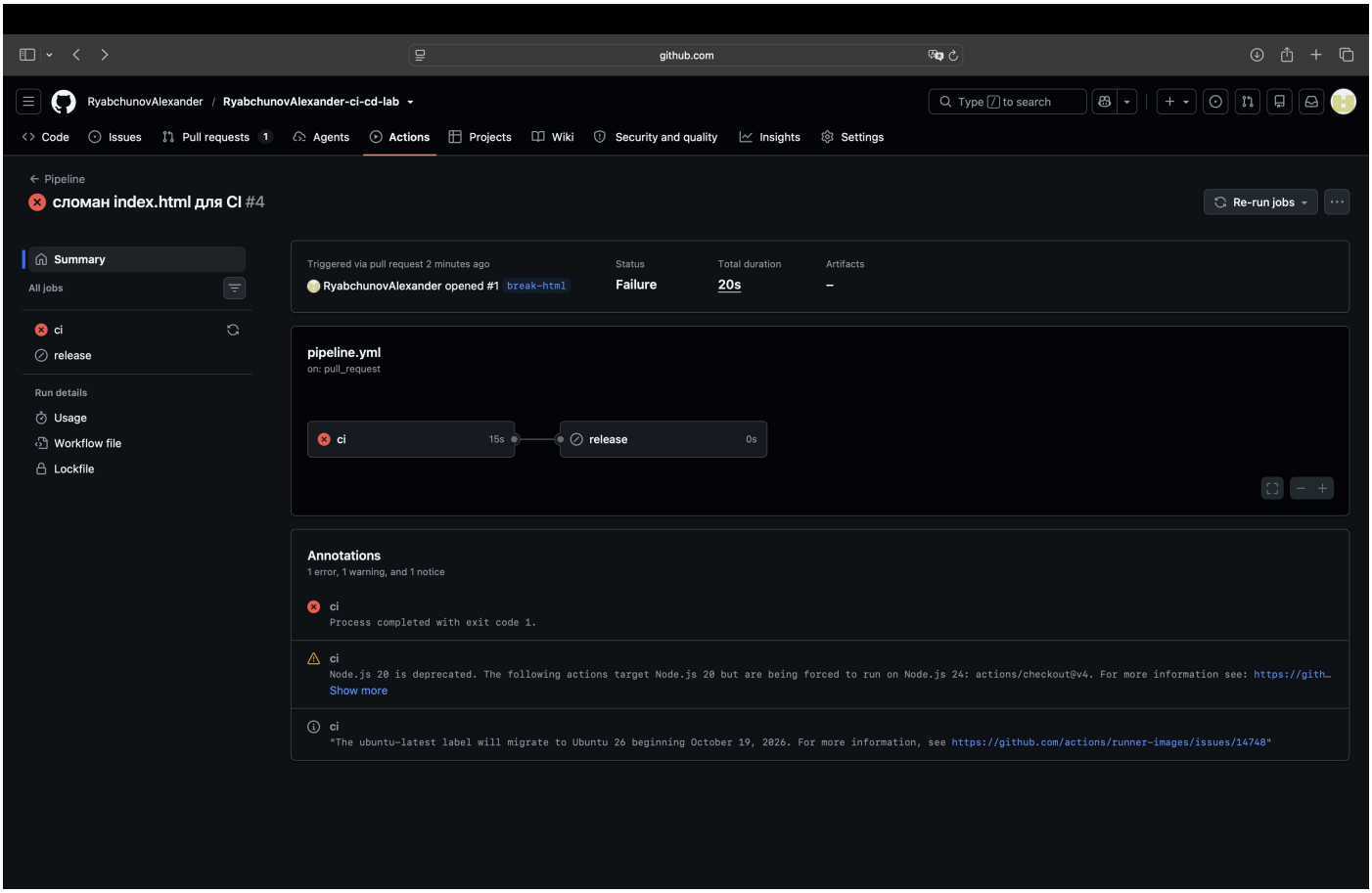
git push origin break-html

```
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % git checkout -b break-html
Switched to a new branch 'break-html'
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % git add index.html
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % git commit -m "сломан index.html для CI"
[break-html 330d440] сломан index.html для CI
 1 file changed, 1 deletion(-)
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab % git push --set-upstream origin break-html
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 317 bytes | 317.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'break-html' on GitHub by visiting:
remote:      https://github.com/RyabchunovAlexander/RyabchunovAlexander-ci-cd-lab/pull/new/break-html
remote:
To https://github.com/RyabchunovAlexander/RyabchunovAlexander-ci-cd-lab.git
 * [new branch]      break-html -> break-html
branch 'break-html' set up to track 'origin/break-html'.
aleksandr@MacBook-Air-Aleksandr RyabchunovAlexander-ci-cd-lab %
```

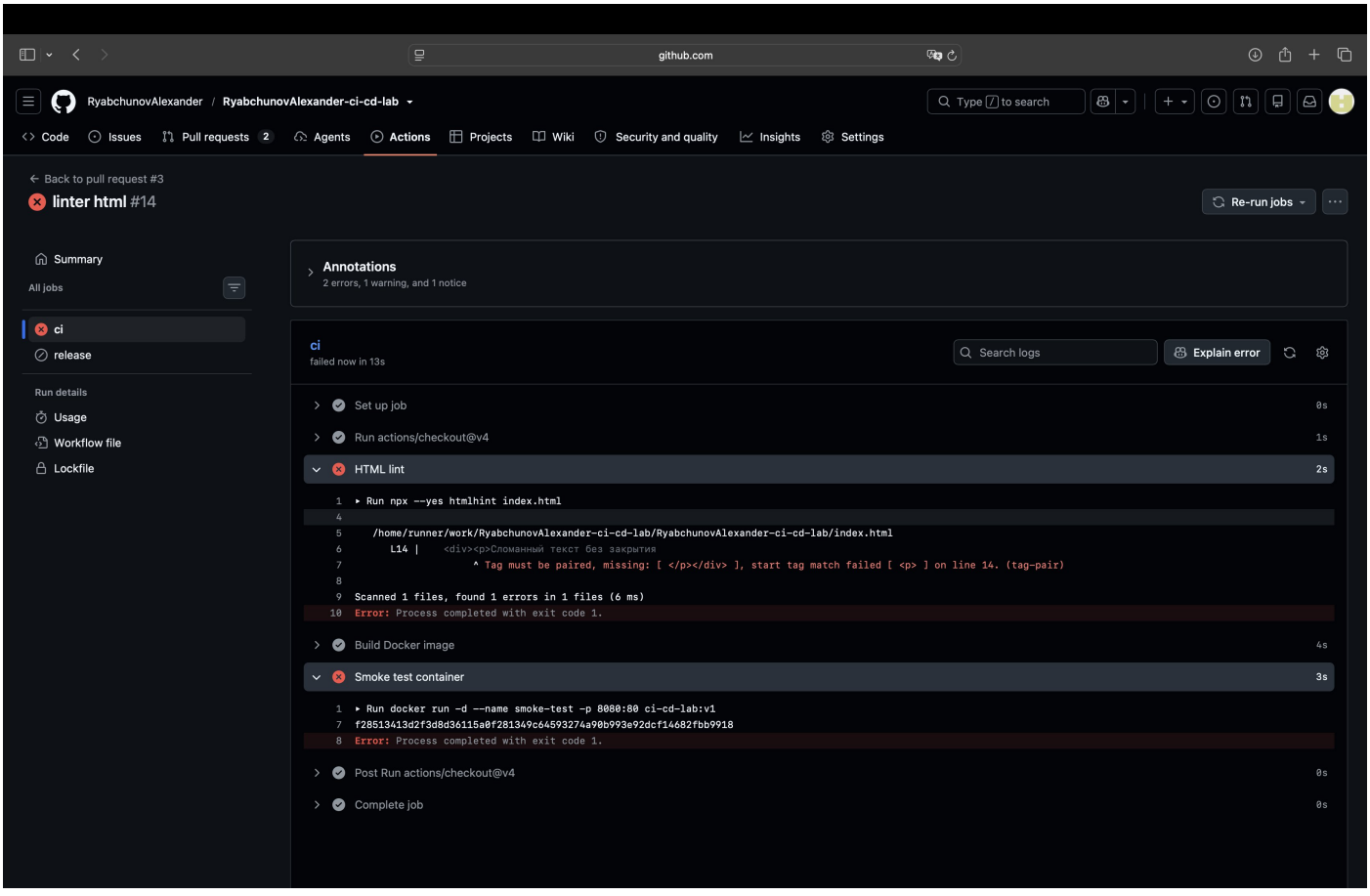
2.3 Анализ поведения CI при создании Pull Request со сломанным кодом.



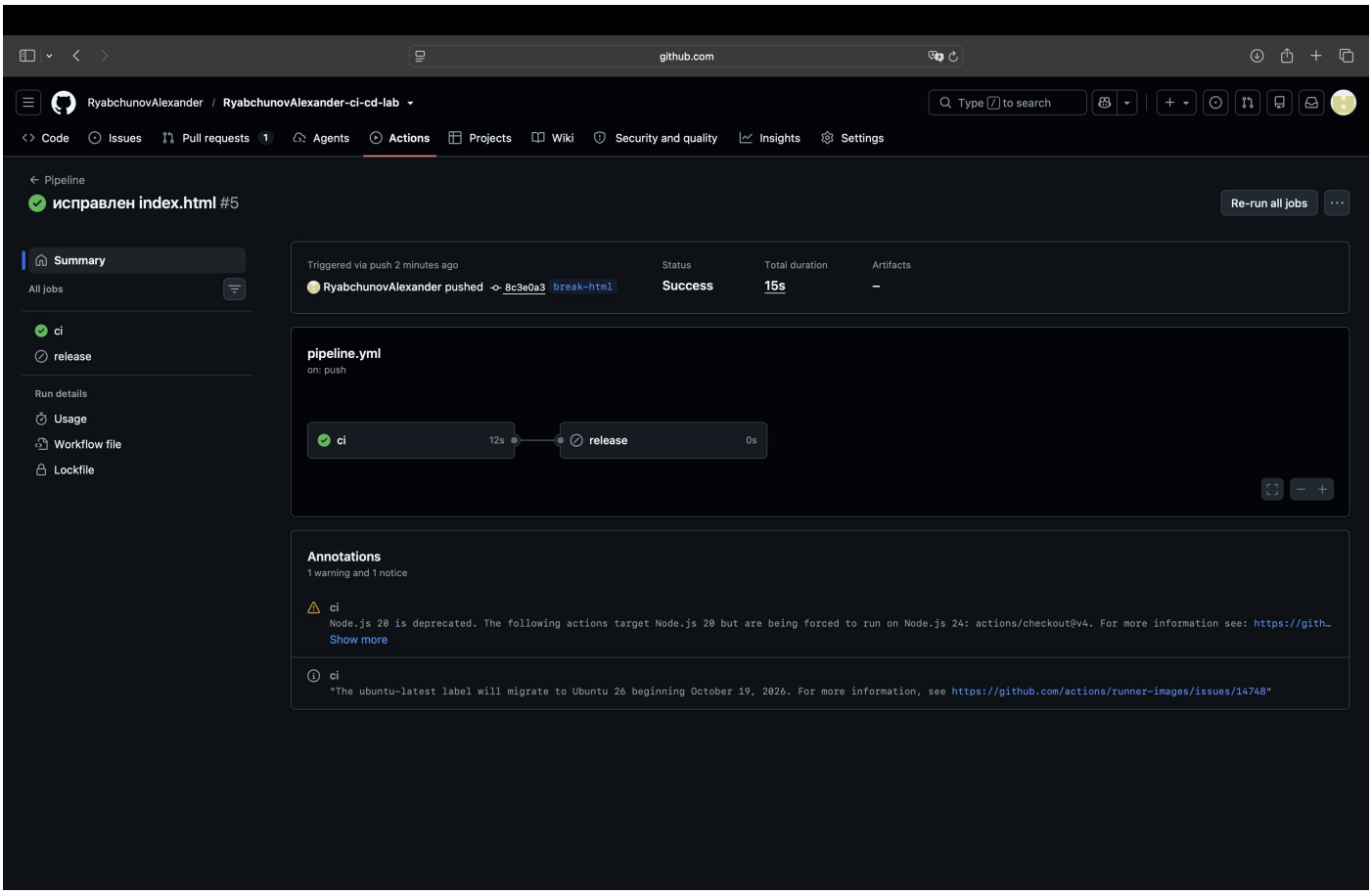
2.4 Проверка работы логических связей needs: ci.



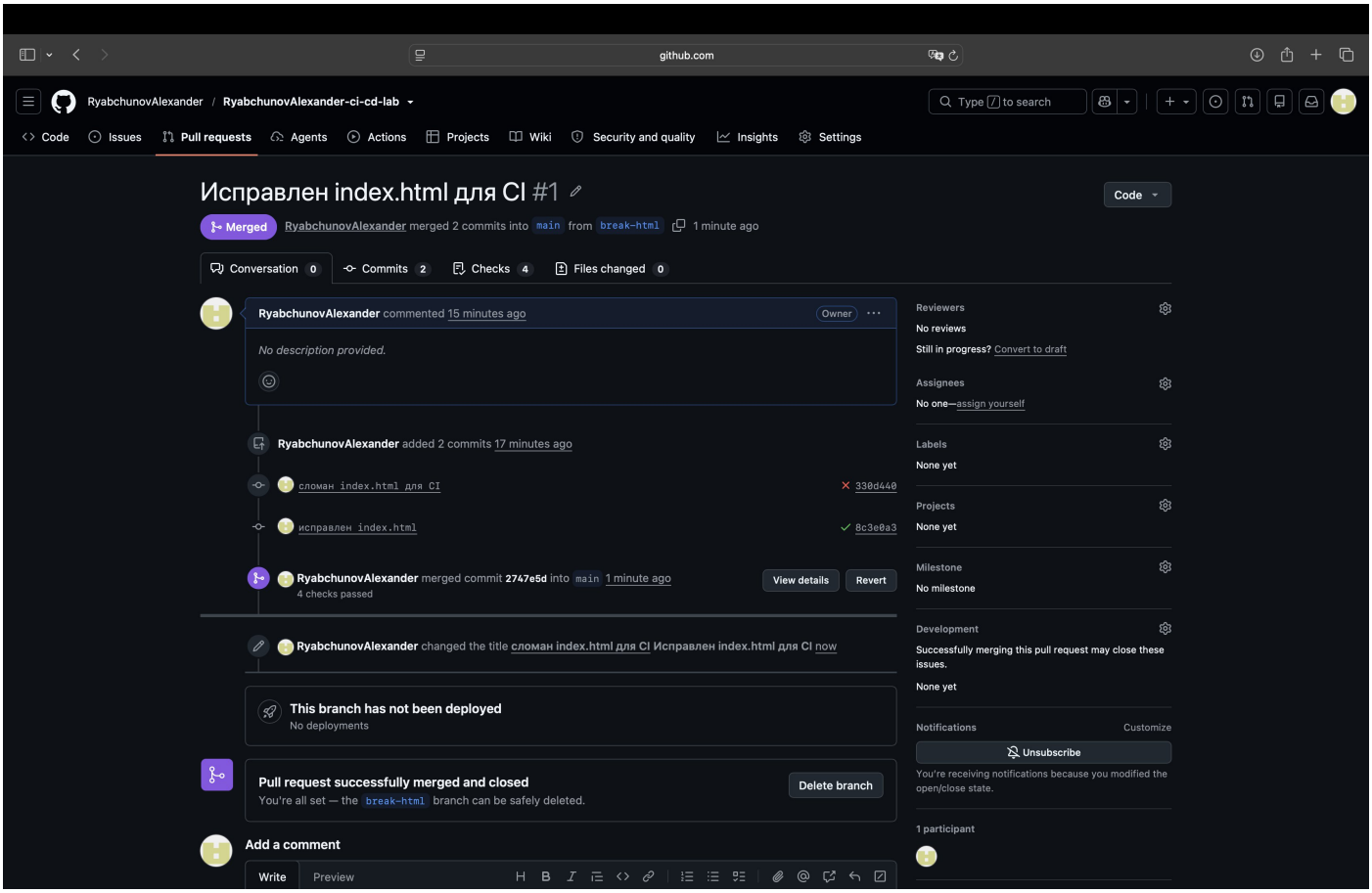
2.5 Четко видна ошибка линтера htmlhint, а последующий шаг Smoke test container завершился с кодом возврата 1, так как тест grep не обнаружил контрольную строку на сломанной странице.



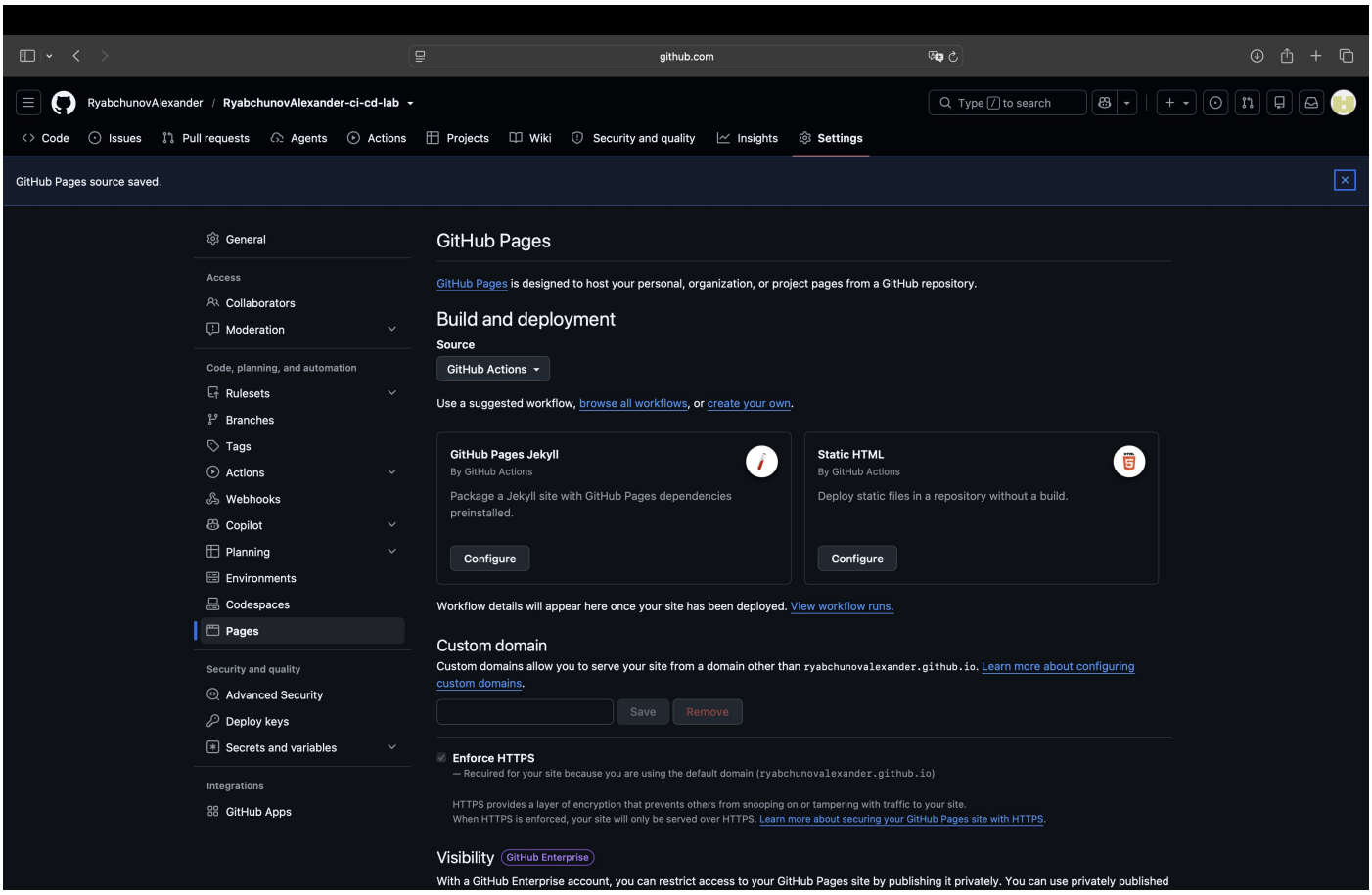
2.6 Отправка исправления в ветку PR. После возвращения корректной структуры HTML и повторного пуша проверка внутри Pull Request успешно прошла и стала зеленой.



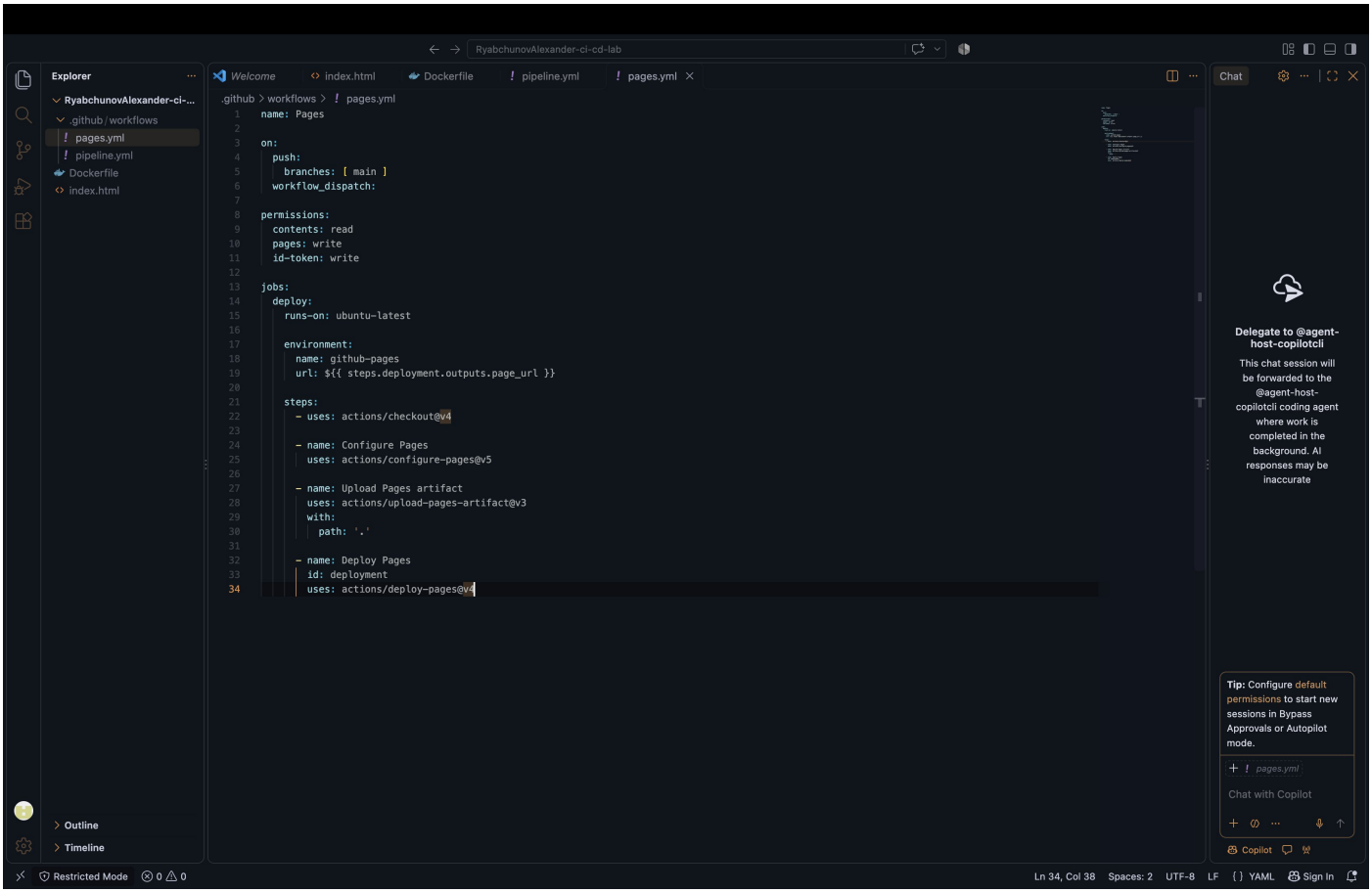
2.7 Слияние проверенной ветки с основной веткой разработки



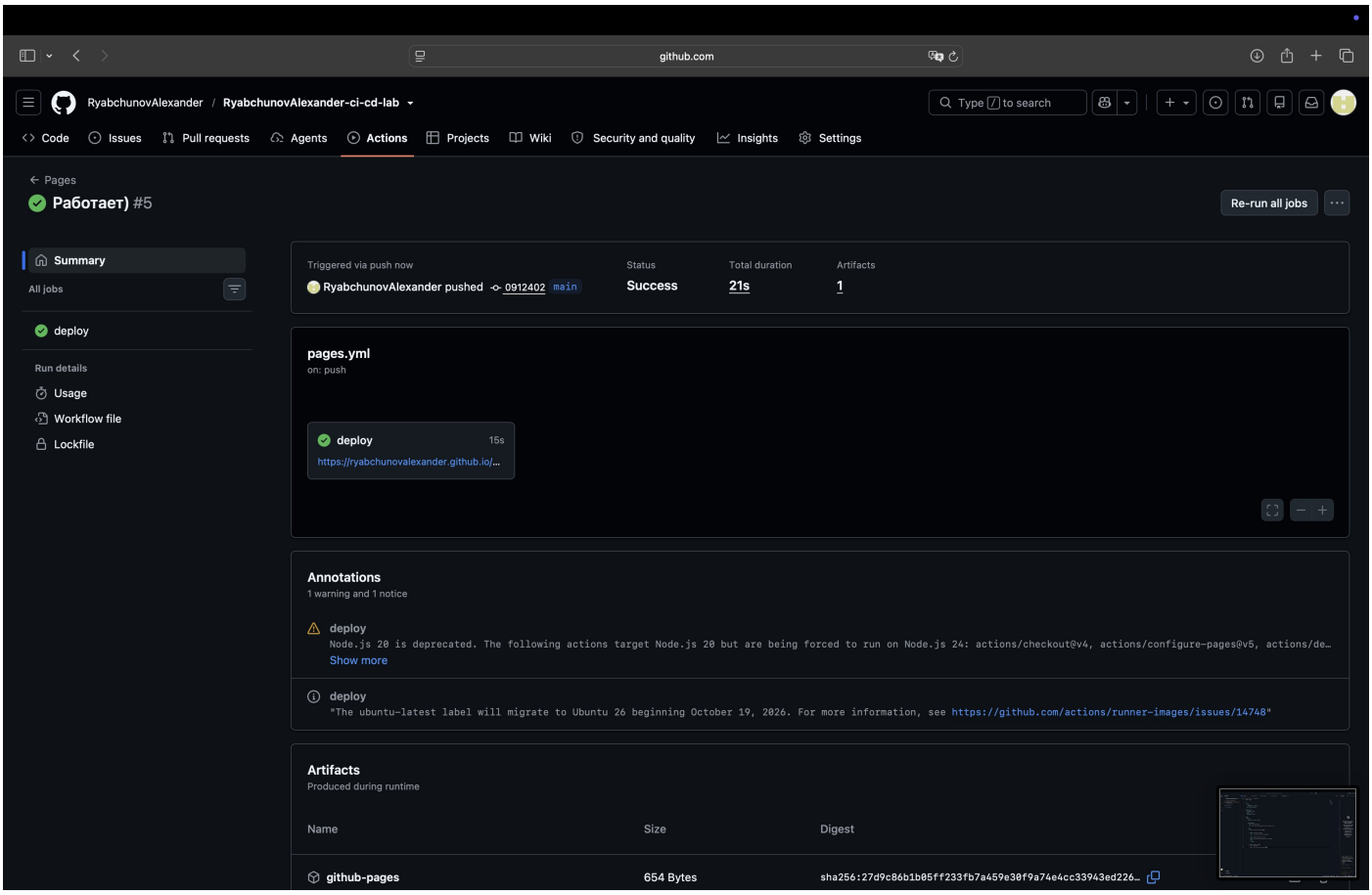
3.1 В меню Settings, Pages в качестве источника (Source) успешно установлен режим GitHub Actions. Отображается плашка готовности сайта.



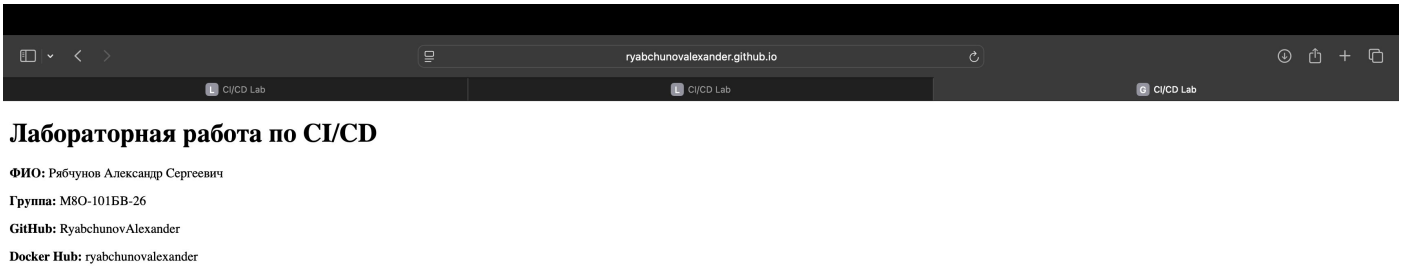
3.2 Демонстрация конфигурационного файла pages.yml



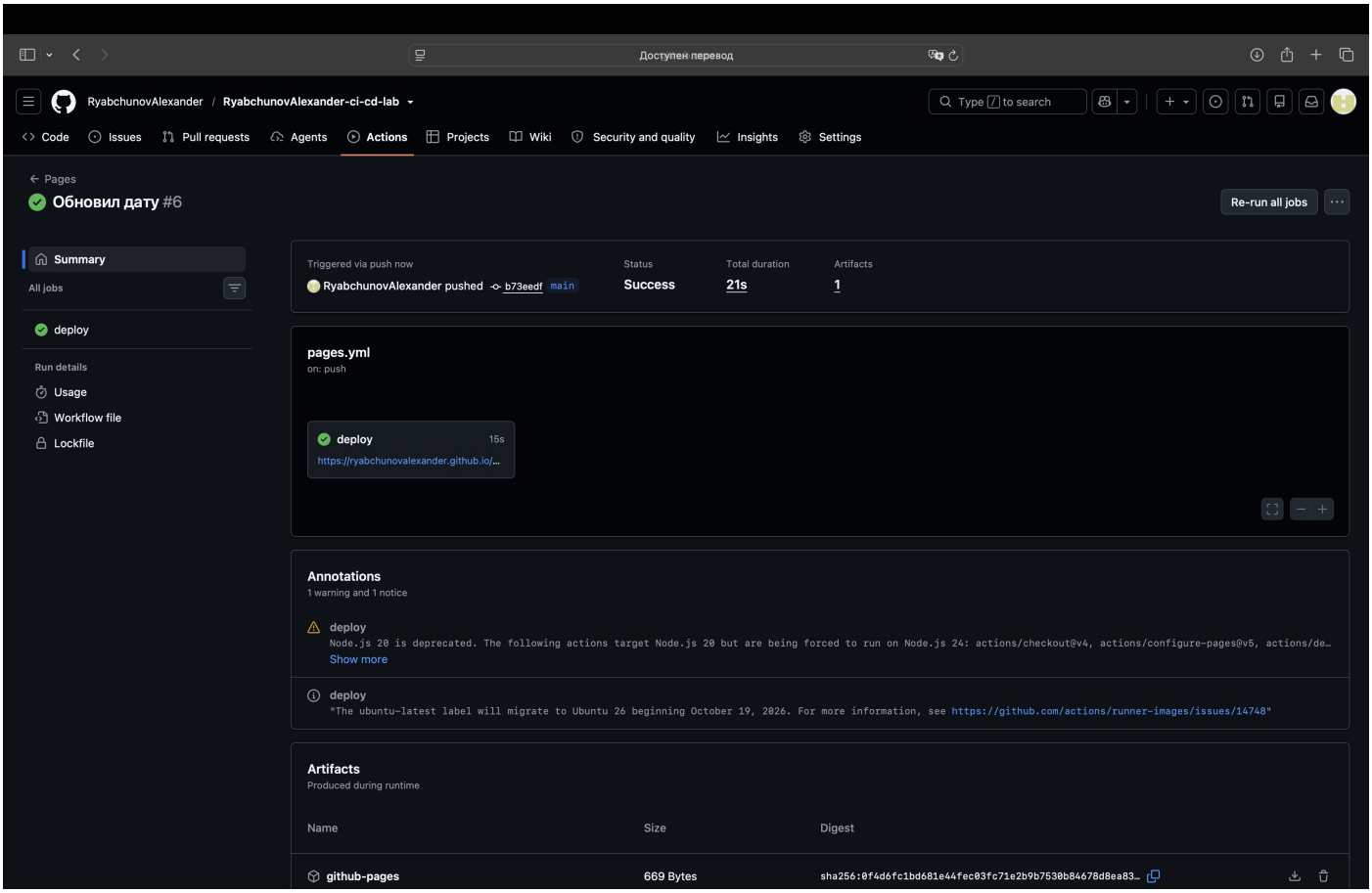
3.3 Мониторинг автоматического деплоя после мерджа. Вкладка Actions показывает успешное выполнение сценария Pages.

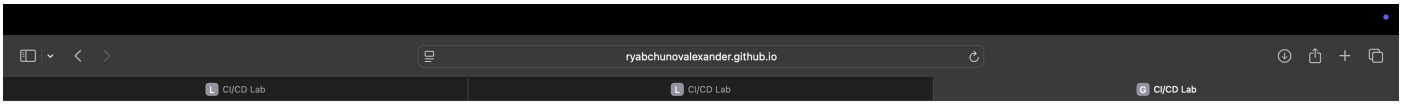


3.4 Проверка доступности сайта в глобальной сети.



3.5 Проверка реактивности системы на обновление контента.





Лабораторная работа по CI/CD

ФИО: Рябчунов Александр Сергеевич

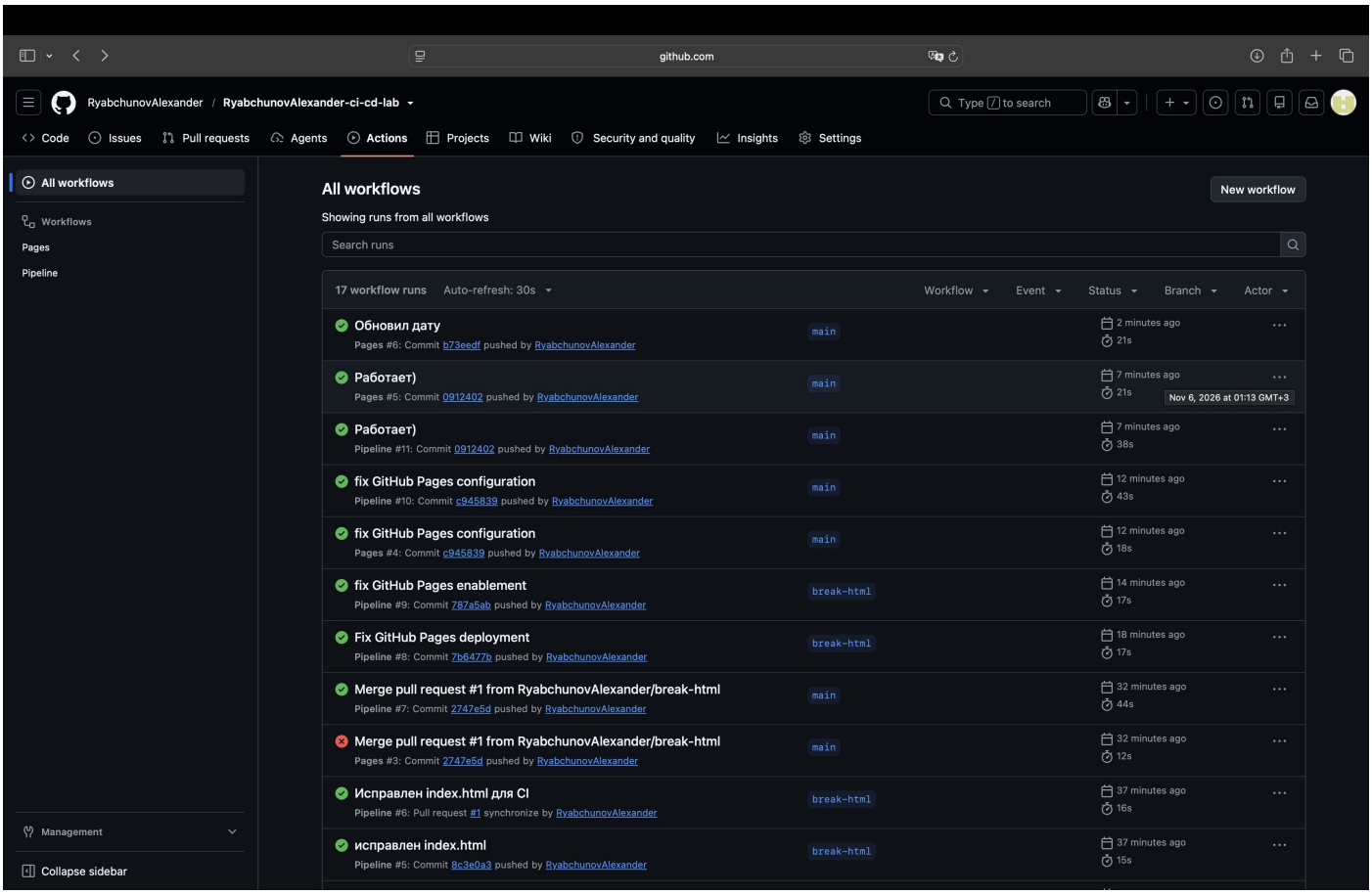
Группа: М8О-101БВ-26

GitHub: RyabchunovAlexander

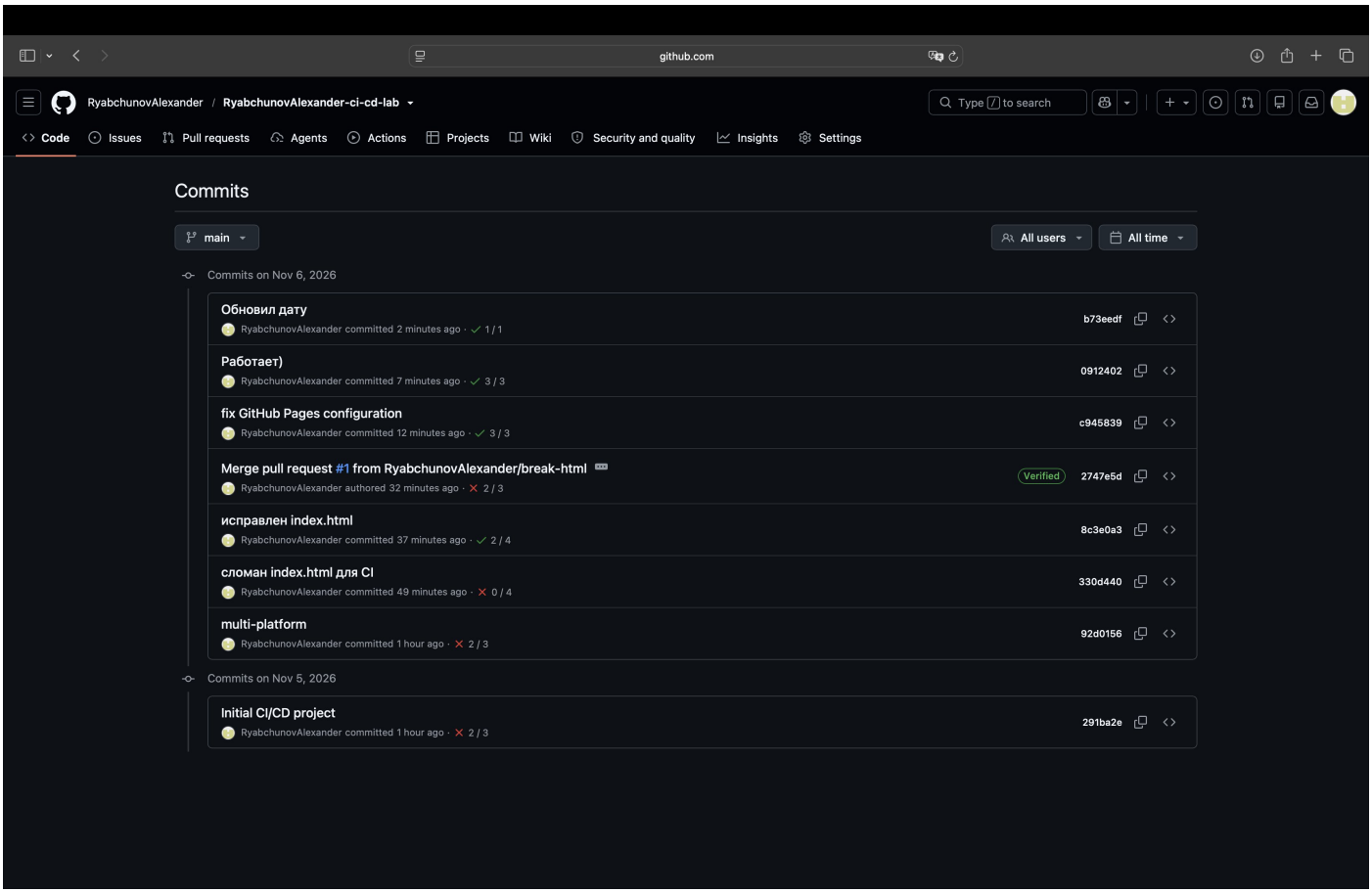
Docker Hub: ryabchunovalexander

Обновлено: 06.11.2026

4.1 Проверка истории автоматических запусков. История во вкладке Actions демонстрирует серию успешных (зеленых) запусков пайплайнов pipeline и pages подряд, сработавших автономно.



4.2 Анализ истории коммитов (Git Commit History). Видна хронологическая цепочка разработки из требуемых коммитов



4.3 Корень проекта на GitHub содержит корректную структуру: веб-страницу, Dockerfile и автоматические сценарии в папке .github/workflows/

